



Aditya Ahlawat

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B.Tech

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🐙 GitHub Profile

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EDUCATION

- **Delhi Technological University** Aug 2023 – Jun 2027
B.Tech in Information Technology CGPA: 8.01
- **Dayawati Modi Academy** 2022
Class XII, Central Board of Secondary Education, Uttar Pradesh Percentage: 94
- **Dayawati Modi Academy** 2020
Class X, Central Board of Secondary Education, Uttar Pradesh Percentage: 95.1

EXPERIENCE

- **FEAT** Jun 2026 – Aug 2026
AI/ML Developer Intern Meerut, India
 - Architected a **RAG chatbot pipeline** with **hybrid search** (BM25 + ChromaDB/Sentence-Transformers) and **cross-encoder reranking** to the top 4 of 10–15 chunks, served by **Llama 3.3 70B** (Groq) with **function-calling** to eliminate hallucinated figures.
 - Implemented an **LLM-as-judge** pipeline to evaluate answer **groundedness** plus a **semantic + keyword search API**, and analysed logged queries via **K-Means** clustering (scikit-learn) to surface FAQ content gaps.
 - Partnered with institute staff to scope requirements, then delivered a **Smart Attendance Management System** (Python, Streamlit, **Supabase/PostgreSQL**) with role-based Admin/Student panels, live on Streamlit Cloud.
 - Added **Haversine anti-proxy verification** via the Geolocation API within a **150 m radius**, shareable deep-linking, and **Pandas-driven** CSV/analytics exports.

PROJECTS

- **AWS Cost Optimizer Dashboard** | 🐙 GitHub May 2026 – Jul 2026
IaC-provisioned serverless stack for AWS spend analytics and LLM-driven cost advisory
 - **Tools & technologies:** Python, AWS (Lambda, API Gateway, Cost Explorer, EC2, RDS, S3), boto3, Terraform, LangChain, OpenAI GPT-4o, Streamlit, Plotly
 - Built a **3-layer serverless application** on AWS using **3 boto3 service modules** to fetch **Cost Explorer** billing data and manage live EC2, RDS, and S3 resources, provisioned end-to-end with **Terraform**.
 - Developed an **AI cost advisor** using **LangChain and GPT-4o**, exposing **4 custom tools** that query real AWS billing and inventory to generate personalized cost-reduction recommendations, surfaced through a **3-page Streamlit dashboard**.
- **TripMate AI** | 🐙 GitHub Mar 2026 – May 2026
A multi-agent AI travel planner that converts a natural-language request into a structured, day-by-day trip plan
 - **Tools & technologies:** Python, MCP, LangGraph, LangChain, FastAPI, Groq (Llama 3), PostgreSQL, Tavily, Docker
 - Architected a **5-agent LangGraph workflow** (flight, hotel, weather, itinerary, response) coordinated through a shared typed state, with **PostgreSQL checkpointing** for stateful multi-turn conversations.
 - Integrated **2 real-time data sources** (AviationStack flights, Tavily web search) and a location resolver supporting **30+ countries and cities**, mapping free-text input to standardized airport codes.
 - Diagnosed LLM hallucination failures (fabricated fares, incoherent multi-region plans) and engineered a **structured-output validation layer** to enforce coherent, source-backed itineraries.

TECHNICAL SKILLS

Languages: Python, SQL, C++

AI & Agentic Frameworks: LangChain, LangGraph, RAG, Model Context Protocol (MCP), LLM APIs (OpenAI GPT-4o, Groq, Llama 3, Ollama)

Web Frameworks & APIs: FastAPI, Pydantic, Streamlit, PostgreSQL, REST APIs

Cloud & Infrastructure: AWS (Lambda, API Gateway, S3, EC2, IAM), Terraform, Docker, Boto3

Developer Tools: Git, GitHub, Cursor, Power BI, Claude Code

Domain Knowledge: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

ACHIEVEMENTS

- **Knight on LeetCode**— Rating **1862**, top **5.81%** globally
- **650+ DSA problems solved**— Across LeetCode, Codeforces, GeeksforGeeks & CodeChef
- **Joint Entrance Examination (JEE)**— Secured Rank **8947** in JEE Advanced & **99.31** percentile in JEE Mains